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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E. Grade Improvement Examinations Nov-2020

Information Science and Engineering

THEORY OF COMPUTATION

Time: 03 Hours ***Maximum Marks: 100***

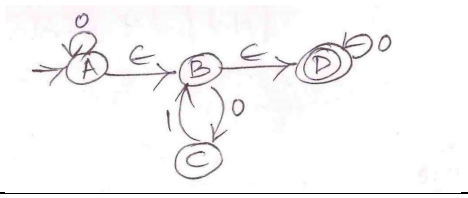
Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART A

1	1.1	Write the transition function of a PDA and NFA.	02
	1.2	Find the string of minimum length in $\{0,1\}^*$ NOT in the language given by $1^*0^*(10)^*$	01
	1.3	Show that the language on $\{a,b\}^*$ having strings that start and end with same character is regular.	02
	1.4	Can there be a DFA without final state? Justify your answer.	02
	1.5	Show that the following grammar is ambiguous . $\sum = \{a,b\}$ $S \rightarrow A\|B\|C$ $A \rightarrow aAb\| \in$ $B \rightarrow bBa\| \in$ $C \rightarrow baC \mid \in$	02
	1.6	Design a turing machine to accept a language having strings of a & b not ending in 'ab'. $\sum = \{a,b\}$	02
	1.7	State pumping lemma for regular languages.	02
	1.8	Identify $L(G)$ for the following production grammar G if productions of 'G' are $S \rightarrow SS$ $S \rightarrow aSb$ $S \rightarrow bSa$ $S \rightarrow \in$	02
	1.9	Show that complement of a recursive language is also recursive.	02
	1.10	Write a CFG to generate strings of odd length having same character at first, middle and last positions. Let $\sum = \{a,b,c\}$	02
	1.11	What is the expansion of Np in Np-complete problems?	01

PART B

2	a	Convert the following $\in -NFA$ to DFA  	06

	b	Show that regular languages are closed under reversal.	04
	c	Show that $\mathcal{L} = \{\omega \epsilon \{1\}^*$ and $\|w\|$ is prime is not a regular language.	06
3	a	Write $CFG's$ to generate i) $L = \{W \in \{a,b,c\}^* \; and \; W \; does \; not \; contain \; substring \; 'abc'\}$ ii) $L = \{W \in \{a,b,c\}^* \; and \; W \; is \; of \; the \; form \; a^i b^j c^k \| j > i + k\}$	06
	b	What is left recursion? Explain how it is removed? Remove left recursion from the grammar whose 'P' is given below $E \rightarrow E + id \; (E - id \; \| id$	06
	c	Convert the following grammar G in to CNF: $S \rightarrow AB$ $A \rightarrow aAb \| \in$ $B \rightarrow Bb \| bb \| \in$	04
		OR	
4	a	Write CFG on $\{e,h\}$ to generate following languages i) $L = \{a^n w w^R b^n; where \; n \geq 1\}$ ii) $L = \{a^m b^n \| \; m \neq n\}$	06
	b	Write a procedure to eliminate useless symbol from the given CFG. Using the procedure eliminate useless symbols from the grammar $S \rightarrow a\text{A} \| \text{bB}$ $A \rightarrow aA \| a$ $B \rightarrow bB$ $D \rightarrow ab \; \| Ea$ $E \rightarrow ac \; \| d$	06
	c	Define null production. Eliminate null productions from the grammar: $S \rightarrow ABC$ $A \rightarrow BC \| a$ $B \rightarrow bAC \| \in$ $C \rightarrow aAB \; \| \in$	04
5	a	State and prove pumping lemma for context free languages.	07
	b	What is a $DPDA$? How do you recognize a PDA as $DPDA$?	03
	c	Obtain a PDA to generate $\mathcal{L} = \{w \| w$ is of the form XX^R (even length palindrome)} on $\sum = \{a,b\}$. Show the sequence of $ID's$ for the string 'abba'	06
		OR	
6	a	Prove or Disprove: $CFL's$ are closed under the following operations. i) Union ii) Concatenation iii) Closure iv) Intersection.	07
	b	Design a PDA to accept the language of all balanced brackets of types {} and []. Trace the moves of the PDA on the string {[]}[]. NOTE: $\sum = \{[,],\{,\}\}$ and the sequence []{, {[}] are invalid	09

7	a	Design Turing machine to accept $L = \{W \mid \eta_a(w) = \eta_b(w)\}$. Trace the moves made by the machine on the string $aabbba$.	08
	b	Write short notes on: i) Halting problem ii) Chomsky Hierarchy.	08
8	a	Define recursive and recursively enumerable languages. Prove that if '$\mathcal{L}$' and its complement are recursively enumerable, then '$\mathcal{L}$' is recursive.	08
	b	Explain the following terms briefly: i) Unsolvable functions ii) Universal TM iii) NP-Completeness iv) Polynomial time reduction.	08